from sklearn.metrics import recall_score, precision_score, f1_score.

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When accuracy is misleading?

Imbalanced dataset:

terrorist or not

N → 999
Y → 1

$$\text{Accuracy} = \frac{999}{999+1} = 99.9\%$$

	1	0
1	0	1
0	0	999

To balance above accuracy, we have precision, recall and F1 Score.

Precision:

Which is better.
Email Classifier:

(1)

Actual	Prediction: Spam	Prediction: Not spam
Spam	100	170
Not spam	30	700

(2)

Actual	Prediction: Spam	Prediction: Not spam
Spam	100	190
Not Spam	10	700

total accuracy of both model = 80%

$FP_1 > FP_2$, $FN_1 < FN_2$

more dangerous is (1) more in spam which is not spam.

So, choose (2).

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$$\text{Precision} = \frac{\text{True Positive (TP)}}{\text{True Positive (TP)} + \text{False Positive (FP)}}$$

(1) (2)

$$\text{Precision} = \frac{100}{100+30} < = \frac{100}{100+10}$$

Recall.

(A)	Detected Cancer	Not Detected	(B)	Detected Cancer	Not Detected.
Has Cancer	1000	200	Has Cancer	1000	500
No Cancer	800	8000	No Cancer	500	8000

which is dangerous, "False Negative"
So, (A) is better.

$$\text{Recall} = \frac{\text{True Positive (TP)}}{\text{True Positive (TP)} + \text{False Negative (FN)}}$$

(A) (B)

$$\text{Recall} = \frac{1000}{1000+200} > \text{Recall} = \frac{1000}{1000+500}$$

If you cannot determine which is dangerous then use.

F1 Score:

- Determine cat or dog.
- Then we do average of both precision and recall → Harmonic mean.

$$F1 \text{ score} = \frac{2 \text{ Precision. Recall}}{\text{Precision} + \text{Recall}}$$

(a) Why Harmonic mean?

Coz it gives priority to lower value. [& penaltilize it more].

a = 5 b = 50

A.M = 27.5

G.M = 15.81

H.M = 9.09

b) a = 80 b = 80

A.M = 80

G.M = 80

H.M = 80

a = 60 b = 100

A.M = 80

G.M = 77.45

H.M = 75

Higher F1 score better score.

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Multi-Class Precision and Recall.

Predicted

Actual	Dog	Cat	Rabbit	Total
Dog	25	5	10	40
Cat	0	30	4	34
Rabbit	4	10	20	34
Total	29	45	34	108

Precision =

- $P_{Dog} = \frac{25}{(25+0+4=29)} = 0.86$
- $P_{Cat} = \frac{30}{45} = 0.66$
- $P_{rabbit} = \frac{20}{34} = 0.58$

Combined Precision:

→ Macro Precision: Arthimetic Avg : $\frac{0.86+0.66+0.58}{3} = 0.70$

balanced dog: 40 cat: 50

when to use

→ Weighted Precision: $\frac{40}{108} \times 0.86 + \frac{34}{108} \times 0.66 + \frac{34}{108} \times 0.58$

$= 0.71.$

imbalanced classes like

Dog	90
Cat	5
rabit	5

when to use.

(7)

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Recall:

$R_{\partial} = \frac{25}{40}$, $R_c = \frac{30}{34}$, $R_R = \frac{20}{34}$

$= 0.62$ $= 0.88$ $= 0.58$

- macro recall : $\frac{0.62 + 0.88 + 0.58}{3} =$
- weighted recall : $\frac{40}{108} \times 0.62 + \frac{34}{108} \times 0.88 + \frac{34}{108} \times 0.58$

F1 score:

$F1_{\partial} = 2 \cdot \frac{P_{\partial} \cdot R_{\partial}}{P_{\partial} + R_{\partial}}$ } other as above.

- precision_score(y_test, y_Pred , average = None)

average = None → separate P_{∂} P_c P_R.
'macro' → avg
'weighted' → weighted.